BIG DATA 101

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WHY BIGDATA

AN INTRODUCTION

‘Always connected and always on’ customers. Generate TBs of data.

HADOOP CAPABILITIES

Types of digital data: structured, semi-structured, unstructured(variety).
Traditional applications problems:

- cannot process semi-structured and unstructured data;
- increase in data size-tera, peta, exa, zeta, yotta;
- how fast the data comes to users-cannot process huge velocity of data;

FB processes 600 TB of data everyday.
Google 100 PB everyday, 2 million searches every second.

4 Vs of Big data:

- Volume
- Velocity
- Variety
- Veracity-accuracy of data

BIG DATA CAPABILITIES

Traditional applications must be able to store, extract huge amounts of data, and scale itself.
2 ways to develop a system/architecture:

- Monolithic- a single system for all processes; performance issue.
- Distributed - machines or nodes; cluster-collection of machines, data and processing distributed; complexity issue;

Hadoop- framework that handles the cluster.
NoSQL - storage mechanism to store different varieties of data;

FAQs

No limit on the number of systems that can be a part of the cluster.
What configuration of machines can be part of the cluster- commodity HW also.
Crashes are a norm in Hadoop.
Are Hadoop and NoSQL together a must-have in the Bigdata environment - No.

Which of the following characteristics of Big Data represents integrity of data used in taking a business decision? Sqoop, Flume.

BIGDATA TECHNOLOGY LANDSCAPE

NoSQL DATABASES

Not only SQL - can store non SQL formats of data as well.
Close to 150+ NoSQL vendors.
Categories:

- Key value store - Redis, Riak
- Column family - Cassandra, HBase
- Document storage - MongoDB, CouchDB
- Graph DB - Neo4J, InfiniteGraph

Introduction to Hadoop Framework

Open source SW framework for distributed storage and parallel processing of big data dataset.
Hadoop distributions: give a customized version; CloudEra, Maphour, CottonWorks;
HDFS - Hadoop distributed file system.
MapReduce - parallel processing; the same program runs on all the does at the same time.
Execution at the node level.

YARN: (yet another resource negotiator) allows various configuration parameters that allow better cluster performance.

Hadoop Framework:



Hadoop Ecosystem

Core part : HDFS, MapReduce, YARN

Data should be uploaded into HDFS first.
MapReduce requires knowledge of Java. Alternatives to it are:

- Hive : by FB; write queries like SQL that then converted into MapReduce; HQL (hive query language).
- Pig : by Yahoo; take in scripts and convert to MapReduce. PigLatin.

Data ingestion tools-

- Sqoop : SQL or Hadoop; allows data transfer from RDBMS source to HDFS, and vice versa;
- Flume : transfer unstructured data into HDFS;

HBase : type of NoSQL;
Ooize : scheduler; workflow management tool.
Zookeeper : service coordination tool.
Mahout : ML applications on HDFS dataset.
Spark : high speed cluster computation framework.

FAQs

Why use MapReduce when there is Pig and HIVE? For complex, java libraries, analysis; for image processing applications.
Hadoop is good for batch processing,not for streaming applications.
Spark for streaming data, real time analysis; also for graph, ML applications.
Can integrate different NoSQLs - replace HBase with Cassandra, Neo4J, etc..

BigData Scenarios

Click stream data analysis
Sentiment analysis- real time, fast analysis
Fraud detection
Risk assessment
Pattern recognition- entertainment industry-recommendation.

Hadoop Architecture

Block size :

Block-smallest unit of storage; default block size: in v1- 64 MB, v2 and beyond-128 MB.
Data size 200 MB : split into 120 mb, 72 mb ; remaining 52 mb wasted.
Block size can be changed.
Small size block: more I/O operations; large block size: wastage of space.

Replication factor:
number of times data is replicated.
Default-3: original, copy1, copy2.
All copies stored in different nodes. Makes Hadoop fault tolerant.

Master salve architecture:

- Name node (master)-has no data, tracks/arbitrates data;
- data nodes : send heartbeat signals.

Secondary name node.
High availability-2 name nodes: active, stand-by node.

HADOOP DEMOS

Unix Basics

The entire HDFS is built on top of the native file system Unix.
Unix is a multiuser OS. Flavors: Linux, SunSolaris.
Features:

- portability- can work on many devices-computers, mobile devices, wearables, etc..
- security
- hierarchical file system: main root directory; traveling across the file system is easy
- redirection tools- redirect output to another file
- pipes-
- background processing functionalities

Unix Demos

pwd : show the current directory
cal : calendar of a specific month
cal 2025 : calendar of the entire year
mkdir usserdir1 : create a directory named userdir1
ls : lists all contents of the current directory
ls -l : long listing; detailed listing of the current directory

(-: file, d: directory)
cd userdir1 : changes directory to userdir1
cat>file1: create a new file named file1 (to create, read, append file)
(then enter text). To come out of the file, ctrl + D
Ctrl + L : clear screen
cat file1 : display file contents
cat >>file1 : append to file
wc file1: word count - number of rows, words, characters.
wc -l file1 : number of lines in the file
cp file1 file2 : copy file1 to file2

Read, write, execute permission on files.

grep unix file1 : search for the word 'unix' in file1
grep -i unix file1 : for case insensitive search (i-ignore case)
head -2 : display top 2 lines of the file
tail -2 : last 2 lines of file
tail -2 file1>file3 : redirect this output to file3 (creates file3 if it doesn't exist)
rm file3 : removes file3
rmdir dir2 : removes directory, provided it is empty
rm -r dir2 : delete directory that has contents (-r: recursive)
man -ls : documentation for commands (man-manual)
q to come out of man
exit : come out of the file

HDFS Demos

hadoop fs -mkdir hdfs_dir1: create a directory named hdfs_dir1
ls : lists the directories present in the native unix, not in HDFS.
hadoop fs -ls : lits hadoop directories
man ls > datafile : in the unix system, 'datafile' is created, having the documentation of ls
hadoop fs -put datafile hdfs_dir1 : put the file 'datafile' from unix to hdfs directory 'hdfs_dir1'
hadoop fs -ls hdfs_dir1 : to check if the file is resent
hadoop fs -cat hdfs_dir1/datafile : display file contents
hadoop fs -get hdfs_dir1/datafile datafile1 : to move file from hdfs back to unix, into file datafile1
hadoop fs -rmdir datafile : to remove directory
hadoop fs -rm file1 : to remove file

Intro to MapReduce Framework

A SW framework to write distributed applications.
Breaks the entire computation into 2 phases: map-execution at node level; reduce-output of map phase aggregated together as key-value pairs.

MapReduce Demos

Intro to Hive

Data warehouse solution built on top of Hadoop. For querying structured data stored on top of HDFS. Has SQL dialect HiveQL. Translates HiveQL to MapReduce prompts.

HIVE	RDBMS
Data managed in HDFS	Data managed by DBS
Schema on read	Schema on write
For working with very large datasets	Smaller datasets
High latency system	Low latency
Not ACID compliant	ACID compliant
Use HiveQL	Uses SQL

HIVE Demos

hive : open the hive terminal
create database hive_db1; : to create a db
show databases : all available DBs
show tables;
use hive_db1; : switches to the selected DB
create table customers(id int, name string, city string);
insert into table customers values(101, ‘sam’,’plano’),(102, ‘aly’,’LA’); : converted into MapReduce job
select * from cust1 limit 5;

MORE ABOUT BIG DATA

Big Data with ML and AI

Big data analytics is mostly used for predictive analytics.
Amazon : flag suspicious transactions.
Meta/FB: user behavioral info

Big Data and Cloud

CC is preferred as the medium for delivery of computing resources.
It is economical to use BD in the cloud.
Provides a flexible infrastructure; as-a-service model hides its complexities.

BD Platforms and Vendors

On-premise:

- Apache Hadoop: open-source;
- CloudEra: 100% open source;
- Hortonworks: enterprise ready, open source; centralized architecture; licensed product;
- MapR: licensed;

Cloud-based (licensed):

- Amazon elastic compute cloud
- Google big data services
- Microsoft Azure for Big data
- IBM Big Data